

Chapter 28: Singleton Breech Delivery — Revision Notes

Williams Obstetrics 26e, pp. 1324–1364 of the PDF. High-yield numbers are in **bold**. The one table (28-1) is transcribed from the PDF image (it is missing from the extracted chapter text). All 13 figures are included (Fig 28-9 is printed twice in the PDF, on p1346 and p1347; one copy is used).

Big picture: Breech persists in **2–5%** of term singletons. Since the **Term Breech Trial**, planned cesarean dominates, but carefully selected women can still deliver vaginally with a skilled provider. **ECV at ≥ 37 wk** is the main way to lower the breech cesarean rate.

1. Classification

Type	Hips	Knees	Lowest part
Frank	Flexed	Extended (feet near the head)	Buttocks
Complete	Flexed (both)	One or both flexed	Buttocks, feet alongside
Incomplete	One or both extended	—	A foot or knee below the breech
Footling (a type of incomplete)	—	—	One or both feet below the breech

- **Stargazing fetus:** extreme neck hyperextension in **~5%** of term breeches. **Flying fetus** = the same with a transverse lie.
 - Look for fetal or uterine anomalies.
 - Vaginal delivery can injure the **cervical spinal cord** → **cesarean indicated** at term. (Cord injury has also followed uneventful cesarean; flexion itself may be to blame.)



Figure 28-1. Frank breech: hips flexed, knees extended, feet by the head.



Figure 28-2. Complete breech: hips and knees flexed, feet beside the buttocks.

2. Diagnosis

Risk factors

- **Preterm** gestation (poles of similar bulk earlier on), **multifetal gestation**.
- Singletons: **extremes of amniotic fluid**, fetal anomalies, **uterine anomalies**, **placenta previa**, nulliparity, ↑ maternal age, **female fetus**, prior breech, **SGA**.
- **Recurrence: 10%** after one breech delivery; **28%** for a third breech.

Examination

- **Leopold:**
 - 1st: hard, round **head in the fundus**
 - 2nd: back on one side, small parts on the other
 - 3rd: soft, movable breech above the inlet (if not engaged)
 - 4th: breech beneath the symphysis after engagement
- Palpation accuracy varies → **sonography** if unclear.
- **Vaginal exam (frank):** ischial tuberosities, sacrum, anus; later the genitalia.
- **Breech vs face** (a swollen breech can mimic a face):

Feature	Breech	Face
Opening	Anus: muscular resistance; finger may be meconium-stained	Mouth: bony jaws and palate
Landmark geometry	Ischial tuberosities + anus in a straight line	Malar eminences + mouth form a triangle

- **Position** is named by the **fetal sacrum**: LSA (back up, sacrum in the maternal left upper/ventral quadrant), RSA, LSP/RSP, LST/RST.

3. Delivery Route

- Factors: parity, pelvic size, pregnancy complications, **provider experience**, patient preference, hospital capability, fetal size, anatomy and GA.

Table 28-1. Factors Favoring Cesarean Delivery of the Breech Fetus

Category	Factors
Clinical characteristics	Lack of operator experience; patient request for cesarean delivery; prior perinatal death or neonatal birth trauma
Sonographic fetal characteristics	Large fetus: >3800 to 4000 g ; severe fetal-growth restriction; term weight <2500 to 2800 g ; oligohydramnios ; fetal anomaly incompatible with vaginal delivery; incomplete breech presentation; hyperextended neck ; apparently healthy, viable preterm fetus either with active labor or with indicated delivery
Maternal characteristics	Pelvic contraction or unfavorable pelvic shape determined clinically or with pelvimetry; prior cesarean delivery

Compiled from text references.

Term breech fetus

Term Breech Trial (Hannah, 2000): 1041 planned cesarean vs 1042 planned vaginal

Outcome	Planned cesarean	Planned vaginal
Perinatal mortality	3/1000	13/1000
"Serious" neonatal morbidity	1.4%	3.8%
Short-term maternal morbidity	Similar	Similar

- Only **57%** of the planned-vaginal group actually delivered vaginally.
- **Criticism:** <10% had radiological pelvimetry; most "serious morbidity" outcomes didn't predict long-term disability. **2-year follow-up: death and neurodevelopmental delay similar.**
- **Supporting cesarean:** WHO (>100,000 deliveries, 9 Asian countries) and others. Meta-analysis absolute risks with vaginal breech: perinatal mortality **0.3%**, birth trauma/neurologic morbidity **0.7%**.
- **Supporting vaginal birth (with strict selection):**
 - **PREMODA** (France, >8000; 2526 selected for planned vaginal, **71%** delivered vaginally): **no difference** in corrected neonatal mortality/outcomes; maternal severe morbidity similar.
 - Lille group: no excess morbidity with strict biometry + pelvimetry.
 - Eide: IQ of >8000 breech-born men unaffected by route.
- US planned vaginal attempts keep falling → **fewer skilled providers**; simulators used for training.
- **Selection criteria for term vaginal breech:**
 - **EFW >2500–2800 g and <3800–4000 g**
 - **No FGR, no oligohydramnios**
 - **Prior cesarean** = rupture risk; limited data suggest worse perinatal outcome (and prior cesarean for arrest suggests a contracted pelvis) → generally avoided
 - **Positive factors:** parity/prior vaginal birth, **spontaneous labor with a normal labor curve**

Preterm breech fetus

- **No RCTs.** Observational data biased (GA groupings, resuscitation thresholds, poor-prognosis fetuses delivered vaginally, incomplete steroids/antibiotics).
- Overall, **planned cesarean confers a survival advantage** (24–32 wk: low vaginal success, ↑ neonatal mortality; meta-analysis >3500 women). One cohort at 26–34 wk found similar outcomes.
- **23–28 wk:** conflicting data.
- Cesarean avoids anoxia, trauma and **head entrapment**, but often needs a **classical incision** (poor lower segment) → ↑ blood loss, infection, **future rupture**.
- **Periviable (20–25⁶/₇ wk) workshop:** data don't consistently support routine cesarean.
- **ACOG/SMFM: consider cesarean from 23⁰/₇ wk; recommend from 25⁰/₇ wk.**

- **32–37 wk**: similar perinatal mortality/morbidity with planned cesarean vs vaginal (>6800).
- → **Individualize** (skill, GA, parity, labor curve, facility). **No resuscitation planned → vaginal delivery likely preferable.**

Delivery complications

- **Maternal:**
 - Genital tract lacerations with either route. At cesarean, forceps or an unmolded head can **extend the hysterotomy**.
 - Vaginally: head through an incompletely dilated cervix or forceps → vaginal/cervical tears, even **uterine rupture**; episiotomy extension, deep tears, cervical incision, infection.
 - Inhalational anesthesia for uterine relaxation → **atony and PPH**.
 - **Maternal death** higher with planned cesarean for breech: **0.47/1000**.
- **Fetal:** prematurity, ↑ **congenital anomalies**, ↑ **cord prolapse**:

Breech type	Cord prolapse (Collea)
Frank	~0.5%
Complete	~5%
Footling	up to 15%

- Rare injuries (more with vaginal birth): humeral/clavicular fracture, brachial plexus injury, **sternocleidomastoid** trauma, epiphyseal separation (scapula, humerus, femur), spinal cord/vertebral injury, genital injury.
- Cesarean does **not** protect against **femur or skull fractures**.
- **Hip dysplasia** is linked to breech **position itself**, not delivery mode.

Memory hook: Cord prolapse "**half, five, fifteen**" = frank, complete, footling.

Imaging

- Sonography confirms presentation and excludes gross anomalies (hydrocephaly, anencephaly), avoiding emergency cesarean for a nonsurvivable fetus.
- ⚠ In preterm or growth-restricted fetuses the **breech is smaller than the aftercoming head**, and the breech head **does not mold**.
- Assess **fetal size, breech type, neck flexion** (two-view abdominal x-ray can show head inclination).
- **Nuchal arm or nuchal cord** on sonography → may warrant cesarean.
- EFW accuracy is **not** altered by breech.
- **BPD >90–100 mm** is often exclusionary.

Pelvimetry (some protocols; Parkland instead relies on **steady labor progress**). CT favored (accurate, low dose, available); MR or plain film also fine.

Measurement	Threshold for planned vaginal birth
Inlet AP diameter	≥10.5 cm
Inlet transverse diameter	≥12.0 cm
Midpelvic interspinous distance	≥10.0 cm
Obstetrical conjugate – BPD	≥15 mm
Inlet transverse – BPD	≥25 mm
Interspinous – BPD	≥0 mm
MR: interspinous >11 cm + true conjugate >12 cm	~75% vaginal success

4. Labor and Delivery

Labor management

- **Team:** (1) provider skilled in breech extraction, (2) assistant, (3) **anesthesia**, (4) newborn resuscitation. **IV access.**
- Don't do pelvimetry if labor is too advanced (risk of delivery in radiology); stepwise progress is itself evidence of pelvic adequacy.
- **One-on-one nursing** (cord prolapse). Continuous EFM preferred; at minimum FHR **every 15 min** in 1st stage. **Scalp electrode on the buttock, avoiding genitalia.**
- **After membrane rupture:** immediate vaginal exam + watch FHR for **5–10 min** (↑ prolapse risk with small/preterm or non-frank breech).
- **Epidural:** may ↑ augmentation and prolong 2nd stage, but gives pain relief and pelvic relaxation for manipulation. Must cover **episiotomy, extraction and Piper forceps**. Nitrous oxide adjunct; general anesthesia must be inducible quickly.

Second stage

Phase	Data / recommendation
Passive (breech descends to the perineum)	Recommended. PREMODA: <30 min in 60%, >60 min in only 20%. Some: cesarean if breech not visible after 1½–2 h
Active (pushing once breech visible)	Hands-off preferred. PREMODA: <30 min in 95% of successes. >20 min ↑ adverse outcome ; Term Breech Trial lowest risk if <30 min ; risk ↑ if >40 min. Canada: cesarean if not delivered/imminent after 60 min of pushing

Induction or augmentation

- Controversial, retrospective data. Meta-analysis: induction → **NICU admission 3% (2× spontaneous labor), cesarean 33%.**
- Breech labor is slower, but steady progress indicates adequate pelvis. Some avoid augmentation; others use it only for hypotonic contractions.
- **Parkland:** amniotomy OK, but **cesarean rather than pharmacologic induction/augmentation.**

Vaginal delivery methods

Method	Definition
Spontaneous breech delivery	Entire fetus expelled with no traction/manipulation , only support
Partial breech extraction	Spontaneous to the umbilicus ; the rest by provider traction and maneuvers
Total breech extraction	Entire body extracted by the provider

Spontaneous breech delivery (cardinal movements)

1. Engagement/descent with the **bitrochanteric diameter in an oblique** diameter; anterior hip descends faster.
2. **Internal rotation 45 degrees** → anterior hip to the pubic arch; bitrochanteric diameter in the AP of the outlet.
3. Anterior hip at the vulva → **lateral flexion** delivers the posterior hip over the perineum → legs follow.
4. Slight **external rotation, back turning anteriorly**; shoulders enter an oblique, then rotate so the **bisacromial diameter is AP.**
5. Head, **sharply flexed**, enters in an oblique, rotates the **nape under the symphysis**, and is **born in flexion.**

Caution

- **Back-posterior rotation is prevented** if possible: the slightest traction can **extend the head** → a longer diameter must pass.

5. Partial Breech Extraction

- Successively **larger, less compressible parts** are born → spontaneous expulsion is the exception.
- **Episiotomy** (mediolateral preferred) unless the perineum is lax.
- Let the breech deliver **spontaneously to the umbilicus**. Once past the introitus, the cord is drawn into the pelvis → **abdomen, thorax, arms and head must follow promptly**.
- Posterior hip usually delivers at **6 o'clock** (often with thick meconium) → anterior hip → external rotation to **sacrum anterior**.

Legs: fingers **parallel to the femur** (avoids femoral fracture); press in the popliteal fossa → knee flexes → deliver the foot; sweep each leg **up and laterally** away from the midline.



Figure 28-3. Delivering a leg: fingers along the femur, popliteal pressure flexes the knee and brings the foot within reach.

Body: thumbs on the sacrum, fingers on the **anterior superior iliac crests** (not the abdomen, to avoid soft-tissue injury). Maternal pushing + downward traction.

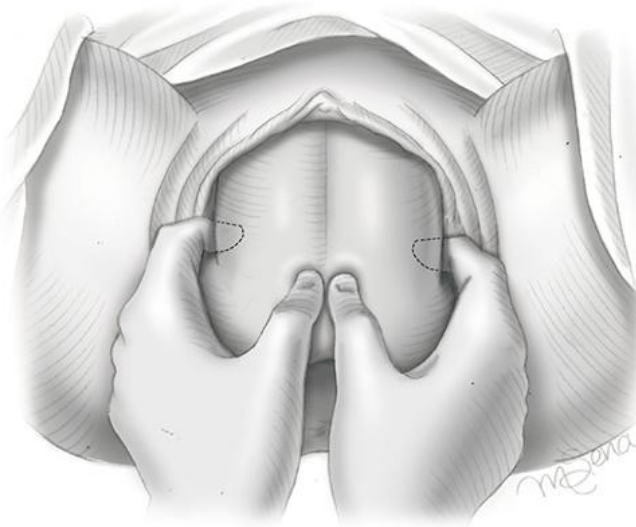


Figure 28-4. Pelvic grip: thumbs over the sacrum, index fingers over the iliac crests; gentle downward traction until the scapulas show.

- **Cardinal rule:** steady, gentle, **downward traction until the lower halves of the scapulas deliver**. Don't attempt the arms until **one axilla is visible**. Either shoulder may be first.

Arms

- **Method 1 (trunk rotation):** rotate the trunk to bring the anterior shoulder into view; **splint the humerus** with fingers parallel (perpendicular force → **humeral fracture**); then rotate **180 degrees the other way** for the second arm.
- **Method 2 (posterior arm first)** if rotation fails: lift the feet up over the mother's inner thigh; hand over the posterior shoulder, sweep the arm up and out over the perineum; then **depress the body** → the anterior shoulder delivers under the pubic arch. The back then turns to the symphysis.

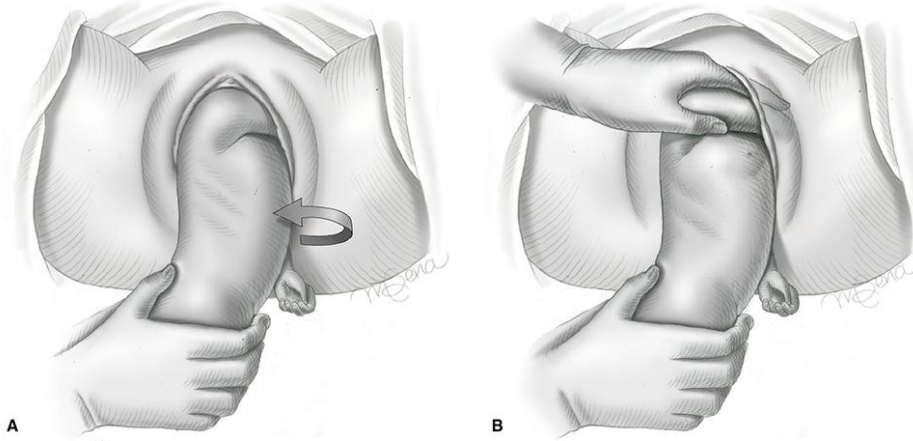


Figure 28-5. After the first arm, a 180-degree rotation (to RST) brings the other shoulder forward; fingers splint the humerus and sweep it across the chest.

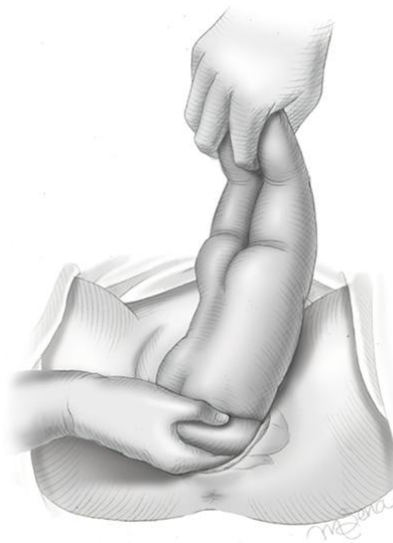


Figure 28-6. Posterior arm first: body lifted over the maternal groin, fingers under the posterior shoulder along the humerus.

Nuchal arm

- Arm(s) trapped **behind the neck** at the inlet.
- **Rotate the fetus through a half circle** so birth-canal friction draws the elbow toward the face:
 - **Right nuchal arm** → **counterclockwise** (back to maternal right)
 - **Left nuchal arm** → **clockwise** (back to maternal left)
- If that fails: push the fetus up to a roomier part of the pelvis and retry, or **extract** the arm (fingers along the humerus, flex the elbow, sweep down the ventral surface). **Humeral or clavicular fracture is common.**

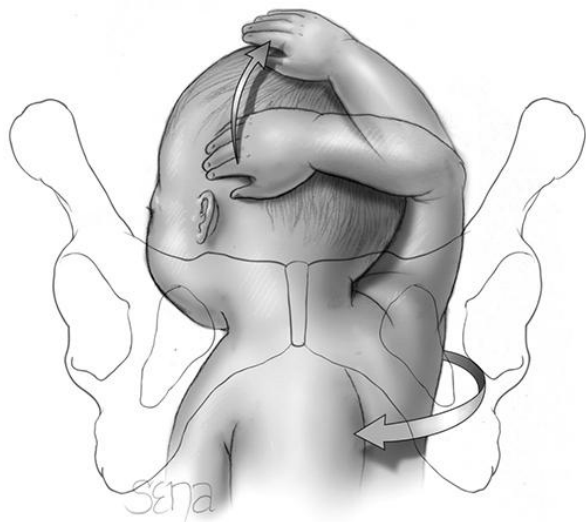


Figure 28-7. Right nuchal arm: rotate the body 180 degrees counterclockwise so friction pulls the elbow toward the face.

Delivery of the aftercoming head (avoid neck hyperextension)

Maneuver	Technique
Mauriceau	Index + middle fingers on the maxilla (flex the head); body on the palm/forearm, legs straddling it. Other hand's two fingers hook over the shoulders . Downward traction until the suboccipital region is under the symphysis; assistant's suprapubic pressure keeps flexion. Then lift the body toward the maternal abdomen → mouth, nose, brow, occiput emerge
Piper forceps (or Laufe-Piper)	Elective or if Mauriceau fails. Apply only once the head is engaged (traction + suprapubic pressure). Body suspended in a towel. Shank has a downward arch and no pelvic curve . Left blade in the left hand for the maternal left side first, then the right. Pull outward while slightly raising the handle; face rolls over the perineum, occiput stays under the symphysis until the brow delivers
Modified Prague	For a back that fails to rotate anteriorly : two fingers grasp the shoulders from below while the other hand draws the feet up over the maternal abdomen

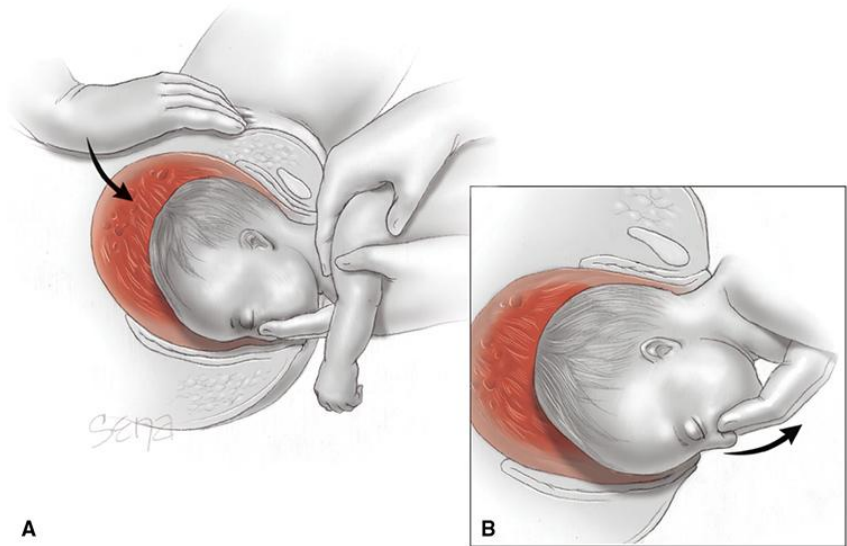


Figure 28-8. Mauriceau maneuver: maxillary pressure plus assistant's suprapubic pressure keeps the head flexed.



Figure 28-9. Piper forceps to the aftercoming head (panel C of 3): body held up in a towel, head delivered in flexion by pulling out while raising the handle.

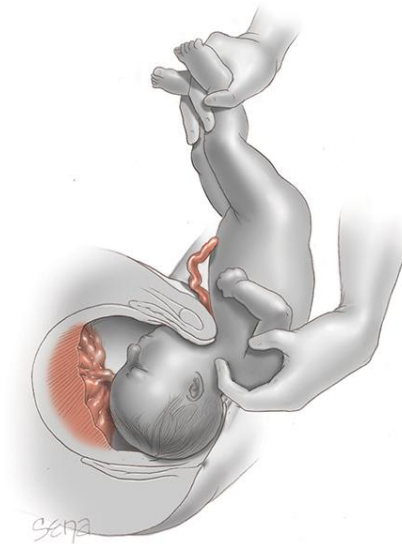


Figure 28-10. Modified Prague maneuver for the back-down fetus.

Head entrapment (a true emergency: assume cord compression)

- **Incompletely dilated cervix** (especially small preterm fetus):
 1. Gentle traction + **manually slip the cervix over the occiput**
 2. **Dührssen incisions** at **2 o'clock**, then **10 o'clock** if needed, rarely **6 o'clock** (avoid the lateral cervical branches of the uterine artery); repair after delivery
 3. **General anesthesia with halogenated agents** to relax the lower segment
 4. **Zavanelli** (replace the fetus, then cesarean)
- **Cephalopelvic disproportion** with arrested head → **Zavanelli or symphysiotomy**.

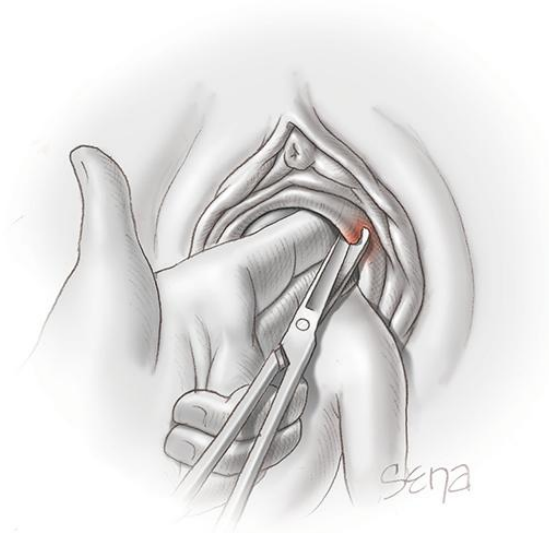


Figure 28-11. Dührssen incision at 2 o'clock (then 10, rarely 6 o'clock) to free a head trapped by the cervix.

6. Total Breech Extraction

- **Frank breech:** moderate traction with **a finger in each groin** + generous episiotomy; then as for partial extraction.
- **Pinard maneuver (decomposition):** converts a frank breech into a **footling** inside the uterus.
 - Easier soon after membranes rupture; very hard with scant fluid and a tight uterus → relax with **general anesthesia, IV magnesium, nitroglycerin or a betamimetic**.
 - Two fingers along a leg press the medial thigh parallel to the femur (externally rotating the hip) + **popliteal pressure** → knee flexes → foot touches the back of the hand → bring it down.
- **Complete/incomplete breech** (when rapid progress prevents cesarean):
 - Hand in the vagina grasps **both feet, middle finger between the ankles**; pull gently; grasp successively higher parts; deliver the hips; thumbs on sacrum, fingers on iliac crests; finish as partial extraction.
 - **Only one foot:** hold it with the **matching hand** (right hand for right foot); other hand follows the leg to find the second foot. If that hip is flexed with the knee extended, hook a finger in the groin.
- The same maneuvers help deliver a breech **through a hysterotomy**.



Figure 28-12. Total breech extraction starts with traction on both feet and ankles.

7. External Cephalic Version (ECV)

- **Version** = substituting one pole for the other or converting oblique/transverse to longitudinal.
 - **ECV**: transabdominal → cephalic.
 - **Internal podalic version**: intrauterine → breech; **reserved for the second twin** (Ch 48).

Indications and timing

- Success **50–60%**; **higher for transverse lie**.
- Performed **before labor at $\geq 37^0/7$ wk**:
 - Balances immaturity vs more fluid earlier.
 - Before 37 wk: $\uparrow$ ECV success but **no $\downarrow$ cesarean** and $\uparrow$ **late-preterm birth**; spontaneous correction still likely; reversion possible.
 - Iatrogenic early-term delivery complications are generally not severe.
- **Absolute contraindication**: vaginal delivery not an option (e.g., **placenta previa**).
- **Relative contraindications**: early labor, **oligohydramnios or ruptured membranes**, known nuchal cord, structural uterine anomaly, **FGR, multifetal gestation**, prior abruption or its risk factors.
- **Prior cesarean**: pooled >100 attempts, no ruptures; Parkland does not attempt.
- **Favorable factors**: **multiparity, unengaged presenting part, nonanterior placenta, nonobese, abundant fluid**. A 2-L IV bolus or amnioinfusion does **not** help. Dedicated breech teams may $\uparrow$ success.

Complications

- Small risks: **abruption, preterm labor, fetal compromise**.
- **Bradycardia is common**; emergency cesarean $\leq 0.5\%$.
- Rare: uterine rupture, **fetomaternal hemorrhage, alloimmunization**, AFE, maternal/fetal death.
- Perinatal morbidity/mortality **not higher** than expectant care.
- After successful ECV, cesarean rates **don't fully fall to vertex baseline** (dystocia, malpresentation, abnormal FHR).

Technique

1. Near facilities for emergency cesarean; **IV access**; **NPO 6 h** (clear liquids until **2 h** before).
2. **Sonography**: confirm nonvertex, adequate fluid, exclude anomalies, placental site, spine orientation.
3. **NST** beforehand (reactivity).
4. **Anti-D immune globulin for Rh D-negative women**.
5. $\pm$ **Tocolysis** and **regional analgesia**.
6. **Left lateral tilt + Trendelenburg**; plenty of gel; monitor fetal heart sonographically.
7. **Forward roll first**: lift the breech out of the pelvis, displace laterally, guide toward the fundus while the head goes to the pelvis. Then **backward flip** if needed.
8. **Stop** for excessive discomfort, persistent abnormal FHR, or multiple failures.

After the attempt

- **After failed ECV**: spontaneous version **7% (2% nulliparas, 13% multiparas)**.
- **After success**: repeat NST until normal.
 - Transient abnormal FHR **6–9%** → IV fluids, O_2 , lateral tilt. Immediate cesarean for tracing only **0.2–0.4%**; a resolved transient tracing does **not** predict later distress.
- **Successful ECV before 39 wk** → **await spontaneous labor** (immediate induction linked to $\uparrow$ cesarean).

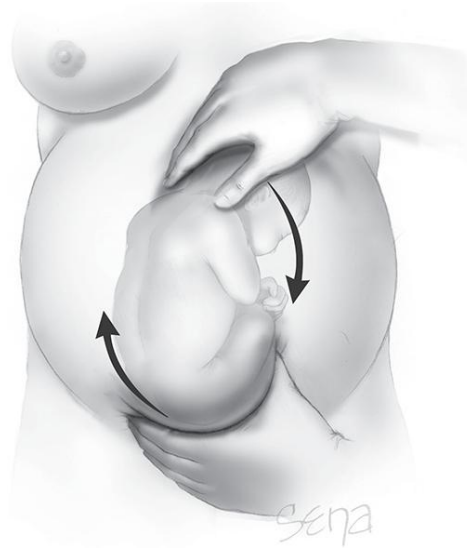


Figure 28-13. ECV forward roll: clockwise pressure on both fetal poles.

Adjuncts

Adjunct	Evidence
Tocolysis (betamimetics)	↑ success. SC terbutaline 52% vs 27% without. Parkland: terbutaline 250 µg SC , start once maternal tachycardia appears. Crosses placenta (mild ↑ FHR baseline)
Other tocolytics	Nifedipine, nitroglycerin, atosiban, salbutamol: limited or unsupportive
Regional analgesia + tocolysis	↑ success vs tocolysis alone; no ↑ FHR changes, emergency cesarean or abruption. Spinal or epidural both reasonable (spinal headache risk)
IV sedation, nitrous oxide	No improvement

Moxibustion

- Burning **mugwort (*Artemisia vulgaris*, moxa)** at acupuncture point **BL 67** to ↑ fetal movement and spontaneous version.
- Done at **33–36 wk** (so ECV is still possible). RCT results **conflicting**.

Quick-Recall: Breech Maneuvers

Problem / step	Maneuver
Frank breech legs out of reach	Pinard (popliteal pressure → knee flexion)
Delivering the body	Thumbs on sacrum , fingers on iliac crests
Arms	Trunk rotation 180° each way (splint humerus), or posterior arm first
Nuchal arm (right)	Rotate counterclockwise
Nuchal arm (left)	Rotate clockwise
Aftercoming head	Mauriceau (maxilla + shoulders + suprapubic pressure)
Aftercoming head (alt.)	Piper forceps (no pelvic curve, head must be engaged)
Back won't rotate anteriorly	Modified Prague
Cervix trapping the head	Dührssen incisions at 2, 10 (6) o'clock; halogenated GA
CPD with trapped head	Zavanelli or symphysiotomy
Breech → cephalic antepartum	ECV ≥37 wk, terbutaline 250 µg SC ± neuraxial
Second twin → breech	Internal podalic version

Key Numbers to Memorize

- Term breech **2–5%**; stargazing **~5%**; recurrence **10%** (second), **28%** (third)
- Term Breech Trial: perinatal mortality **3 vs 13/1000**; serious morbidity **1.4 vs 3.8%**; only **57%** delivered vaginally
- PREMODA: **71%** vaginal success in selected women
- Vaginal candidate EFW **2500–2800 to 3800–4000 g**; BPD **>90–100 mm** excludes
- Pelvimetry: inlet AP **≥10.5**, transverse **≥12.0**, interspinous **≥10.0 cm**
- Cord prolapse: frank **0.5%**, complete **5%**, footling **15%**
- Maternal death with planned breech cesarean **0.47/1000**
- Periviable: consider cesarean from **23⁰/7**, recommend from **25⁰/7 wk**
- FHR every **15 min** (minimum); watch **5–10 min** after membrane rupture
- Passive 2nd stage: cesarean if breech not visible by **1½–2 h**; active pushing ideally **<30 min**, cesarean after **60 min**
- Induction: NICU **3%**, cesarean **33%**
- ECV at **≥37 wk**; success **50–60%**; terbutaline **250 µg SC (52% vs 27%)**; emergency cesarean **≤0.5%**; transient abnormal FHR **6–9%**; spontaneous version after failure **7%**
- NPO **6 h** (clear liquids until **2 h**); moxibustion at **33–36 wk**